

# Homework #3

## – Simple DL Workflow in Kubeflow

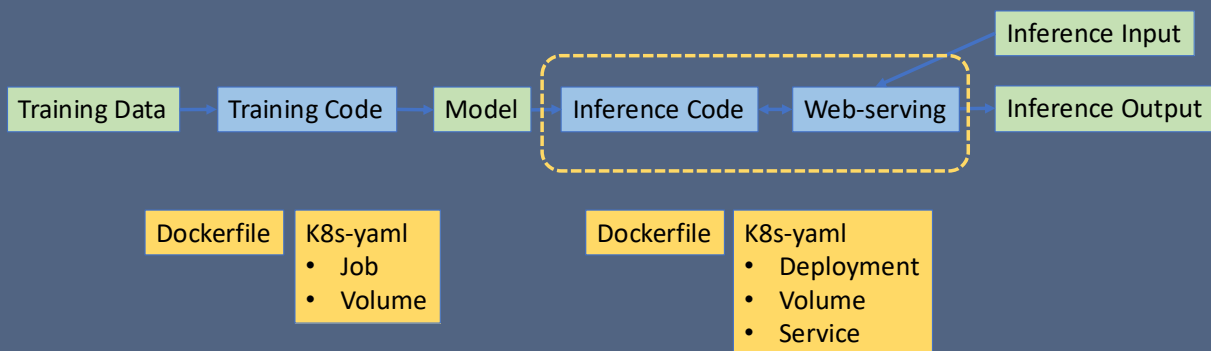
- Objective:
  - Develop Container and Kubernetes artifacts perform DL training and DL inference hosting in GKE: Google Kubernetes Engine.
  - The work demands self-driven deep dive into Kubernetes, Kubeflow concepts and components.
- Submission:
  - Dockerfile, K8S yaml files, instructions on how to build and run the service on K8S clusters on GCP-GKE. Screen captures on how the app run.
  - Document your work and report on your experiences: e.g.
    - What Kuberentes controllers did you use for training and inference and why?

## Homework #3

### – Architecture Aspect

- Application containers
  - Training: training program, container definition file (dockerfile), training job yaml file
  - Inference: inference program, dockerfile, service yaml file, deployment yaml file
    - Training and inference program need to store and load the trained model persistently.
    - Host a URL to allow interaction with User.
- User interaction
  - Give any input to the inference engine (a number, a space bar, image file, image id in the data set) to show that the inference is happening in an interactive/controlled manner.
- Persistent storage (PVC) – if required (optional)
  - After creating your cluster, you can create a PVC resource:
    - GCP → Kubernetes Engine → Storage → Persistent storage volume

# The Abstracted Workflow



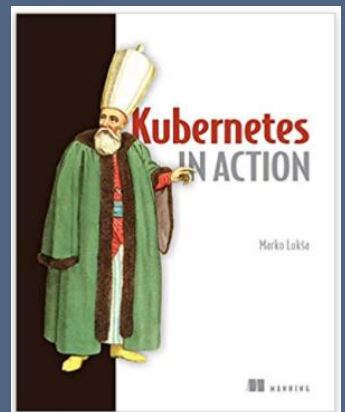
| Rubric              |         |                                                                                                                            | Grading       |                    |
|---------------------|---------|----------------------------------------------------------------------------------------------------------------------------|---------------|--------------------|
| Category            | Partial | Deduction rules                                                                                                            | partial score | Deduction comments |
| Created k8s cluster | 40      | deduct 4 for not providing reasonable amount of proof of work.                                                             |               |                    |
| Training in k8s     | 30      | deduct 2 for missing any of the steps: dockerfile, docker push, train-yaml, model saving                                   |               |                    |
| Inferencing in k8s  | 25      | deduct 2 for missing any of the steps: PVC creation and preparation, code augmentation, dockerfile, infer-yaml, infer-test |               |                    |
| writing             | 5       | up to 2 point deduction if lack of attention to have readers to follow.                                                    |               |                    |
| late charge         |         |                                                                                                                            |               |                    |
| TOTAL SCORE         | 100     |                                                                                                                            |               |                    |

# Example on how to add a GPU to K8S Cluster

```
apiVersion: v1
kind: Pod
metadata:
  name: myapp-pod
  labels:
    app: myapp
spec:
  containers:
  - name: myapp-container
    image: busybox
    command: ['sh', '-c', 'echo Hello Kubernetes! && sleep 36']
  resources:
    limits:
      nvidia.com/gpu: 1
  nodeSelector:
    cloud.google.com/gke-accelerator: nvidia-tesla-t4
```

# Suggested Study Material

- Borg, Omega, and Kubernetes:
  - <https://ai.google/research/pubs/pub44843>
- Introduction to Kubernetes:
  - <https://www.digitalocean.com/community/tutorials/an-introduction-to-kubernetes>
- Facebook Tupperware:
  - <https://engineering.fb.com/data-center-engineering/tupperware/>
- Kubernetes deconstructed:
  - <https://www.youtube.com/watch?v=90kZRYpCRZw>
  - <http://kube-decon.carson-anderson.com/Layers/0-Intro.sozi.html#frame5378>
- The History of Kubernetes on a Timeline:
  - <https://blog.risingstack.com/the-history-of-kubernetes/>
- The State of the Kubernetes Ecosystem [eBook]:
  - <https://thenewstack.io/ebooks/kubernetes/state-of-kubernetes-ecosystem/>
- Kubernetes In Action by Marko Luksa [Book]



## Lab

- Try the various example files here:
  - <https://github.com/seelam/k8s>